



MATEMATICI 2

Curs 17

ex.:

$$y_1' = y_2$$
$$y_2' = y_1 + e^x$$
$$y_1(0) = 1$$
$$y_2(0) = 2$$

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}' = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} + \begin{pmatrix} 0 \\ e^x \end{pmatrix}$$

$$y' = A \cdot y + b(x)$$

$$(*)\begin{cases} x_0=0,\quad y_0=\begin{pmatrix}1\\1\end{pmatrix} \\ y(x)=\begin{pmatrix}y_1(x)\\y_2(x)\end{pmatrix}=e^{xA}\left(e^{-x_0A}\cdot y_0+\int_{x_0}^{x} e^{-\tau A}\cdot b(\tau)\,d\tau\right) \end{cases}$$

$$\det(A-\lambda I_2)=0 \Rightarrow \begin{vmatrix}-\lambda & 1\\ 1 & -\lambda\end{vmatrix}=0 \Rightarrow \lambda^2-1=0 \Rightarrow \begin{cases}\lambda_1=1\\ \lambda_2=-1\end{cases}$$

$$V_{\lambda_1}=\left\{\begin{pmatrix}x\\y\end{pmatrix}\in\mathbb{R}^2 \;\middle|\; (A-\lambda_1 I_2)\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}\right\}$$

$$\begin{pmatrix}-1 & 1\\ 1 & -1\end{pmatrix}\cdot\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix} \Rightarrow \begin{matrix}-x+y=0\\ x-y=0\end{matrix}\Rightarrow y=x=\alpha \Rightarrow$$

$$\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \alpha \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$V_{\lambda_1} = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$ = bază pt. V_{λ_1}

$(A - \lambda_2 I_2) \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow y + x = 0 \Rightarrow x = \beta$

$y = -\beta \quad \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \beta \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$V_{\lambda_2} = \left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$ = bază pt. V_{λ_2}

$$B = \{v_1, v_2\}$$

$$T = \{v_1 \mid v_2\} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$T^{-1} \cdot A \cdot T = D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$A = T \cdot D \cdot T^{-1}$$

$$t \cdot A = T \cdot (t \cdot D) \cdot T^{-1}$$

$$e^{tA} = T \cdot e^{tD} T^{-1} = T \begin{pmatrix} e^{1 \cdot t} & 0 \\ 0 & e^{-1 \cdot t} \end{pmatrix} T^{-1}$$

$$\begin{cases} X_1 + X_2 = y_1 \\ X_1 - X_2 = y_2 \end{cases}$$

$$x_1 = \frac{1}{2} y_1 + \frac{1}{2} y_2$$

$$x_2 = \frac{1}{2} y_1 - \frac{1}{2} y_2 \quad \Rightarrow T^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$$

$$A \cdot \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \Rightarrow \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = A^{-1} \cdot \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$e^{tA} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} e^t & 0 \\ 0 & \frac{1}{e^t} \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} e^t & e^{-t} \\ e^t & -e^{-t} \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$$

$$\Rightarrow e^{tA} = \begin{pmatrix} \frac{e^t}{2} + \frac{e^{-t}}{2} & \frac{e^t}{2} - \frac{e^{-t}}{2} \\ \frac{e^t}{2} - \frac{e^{-t}}{2} & \frac{e^t}{2} + \frac{e^{-t}}{2} \end{pmatrix}$$

$$y(x)=\begin{pmatrix} \frac{e^x}{2}+\frac{e^{-x}}{2} & \frac{e^x}{2}-\frac{e^{-x}}{2} \\ \frac{e^x}{2}-\frac{e^{-x}}{2} & \frac{e^x}{2}+\frac{e^{-x}}{2} \end{pmatrix}\cdot\left[\begin{pmatrix} \frac{1}{2}+\frac{1}{2} & \frac{1}{2}-\frac{1}{2} \\ \frac{1}{2}-\frac{1}{2} & \frac{1}{2}+\frac{1}{2} \end{pmatrix}\cdot\begin{pmatrix}1\\1\end{pmatrix}+\right.$$

$$\left.+\int_0^x \begin{pmatrix} \frac{e^{-\tau}}{2}+\frac{e^{\tau}}{2} & \frac{e^{-\tau}}{2}-\frac{e^{\tau}}{2} \\ \frac{e^{-\tau}}{2}-\frac{e^{\tau}}{2} & \frac{e^{-\tau}}{2}+\frac{e^{\tau}}{2} \end{pmatrix}\cdot\begin{pmatrix}0\\e^{\tau}\end{pmatrix}d\tau\right]$$

de continuat – temă.

sau $e^{xA} = Y(x)$ (y mare)

Ecuații diferențiale de ordin superior :

- o ec. dif. de ord. n are forma generală

$$x^{(n)} = F(t, x, x', \dots, x^{(n-1)}) \; ; \; F: U \subset \mathbb{R}^{n+1} \to \mathbb{R} \; ; \; F \text{ cont}$$

$\varphi: I \to \mathbb{R}$, φ de cls. $C^n(i)$ este sol. dacă:

1) $(t, \varphi(t), \varphi'(t), \dots, \varphi^{(n-1)}(t)) \in U, \; \forall t \in I)$

2) $\varphi^{(n)}(t) = F(t, \varphi(t), \varphi'(t), \dots, \varphi^{(n-1)}(t)), \; \forall t \in I.$

Teoremă :

Ecuația dif. (1) este echiv. cu sistemul de ec. dif :

$$2)\begin{cases} x_1' = x_2 \\ x_2' = x_3 \\ \vdots \\ x_{n-1}' = x_n \\ x_n' = F(t, x_1, \dots, x_n) \end{cases}$$

f sol. a ec. 1) $\Leftrightarrow$ $f_1 = f$, $f_2 = f'$, ..., $f_n = f^{(n-1)}$ sol. sist 2)

$f(t_0) = u_1, \ldots, f^{(n-1)}(t_0) = u_n \Leftrightarrow f_1(t_0) = u_1, \ldots, f_n(t_0) = u_n$.

Cp. de ec. dif. de ordin superior:

a) $y^{(n)} = f(x)$; $f: I \to \mathbb{R}$ cont.

b) $F(x, y^{(k)}, y^{(k+1)}, \ldots, y^{(n)}) = 0$

$y^{(k)}(x) = z(x)$

$y^{(k+1)}(x) = z'(x)$

$\vdots$

c) Ec. omogene în $y, y', \dots, y^{(n)}$:

Se reduce ord. cu 1 un. cu substitutia $z(x) = \frac{y'(x)}{y(x)}$

d) ec. autonome (nu apare x):

$y' = P(y)$

$\dot{} = \text{derivare în funcție de } y.$

$$y'' = \frac{dy'}{dx} = \frac{dP}{dx} = \frac{dP}{dy}\cdot\frac{dy}{dx} = \dot{P}\cdot y' = \dot{P}\cdot P$$

$$y''' = \frac{dy''}{dx} = \frac{d(\dot{P}\cdot P)}{dx} = \frac{d(\dot{P}\cdot P)}{dy}\cdot\frac{dy}{dx} = (\ddot{P}\cdot P + \dot{P}\cdot\dot{P})\,P$$

- se reduce ord. ec. cu 1 unitate

e) ec. dif. liniare de ordinul n

$$a_0(t)\cdot x^{(n)}(t) + a_1(t)\, x^{(n-1)}(t) + \dots + a_n(t)\, x(t) = b(t)$$

unde $a_i, b : \underset{\hat{\mathbb{R}}}{I} \to \mathbb{R}$, $i = \overline{0,n}$, $a_n \neq 0$

$b(t) = 0 \Rightarrow$ ec. liniară de ord. n omogenă

$$S = \{ x : I \to \mathbb{R} \mid \mathcal{L}(x(t)) = 0 \}$$

S – s.v. real cu $\dim = n$.

$$\mathcal{Y} = \{ x : I \to \mathbb{R} \mid \mathcal{L}(x(t)) = b(t) \}$$ pt. $\forall x_p$ sol. particulară a ec. neomogene

$\mathcal{Y}$ = mult. sol. ec. dif. <u>neomogene</u>.

$$\mathcal{Y} = x_p + S = \{ x_p + x \mid x \in S \}$$

Def: orice bază din S s.n. sistem fundamental de soluții. Orice sol. din S - comb. liniară a sol. particulare.

Dacă $\{y_1, \ldots, y_n\}$ = sist. fundamental de sol. $\rightarrow$ sist.

următorul asociat (2) este :

$$W(t)=\begin{vmatrix} f_1 & f_2 & \dots & f_n \\ f_1' & f_2' & \dots & f_n' \\ \vdots & & & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \dots & f_n^{(n-1)} \end{vmatrix} \xrightarrow[\text{Liouville}]{\text{Th. lui}} W(t)=W(t_0)\cdot e^{\int_{t_0}^{t}\frac{a_1(\tau)}{a_0(\tau)}d\tau}.$$

- în pract., fol. met. var. ct. a lui Lagrange, dacă $f_1,\dots,f_n$ = sist. fund. de sol. at. ec. sist. desc.: $x_0=c_1f_1+\dots+c_nf_n$

- o sol. part. a sist. omogen: $x_1=c_1(t)f_1+\dots+c_n(t)f_n$

$c_1,\dots,c_n\in\mathbb{R}$.

$$\begin{cases} c_1' y_1 + \dots + c_n' y_n = 0 \\ c_1' y_1' + \dots + c_n' y_n' = 0 \\ \vdots \\ c_1' \cdot y_1^{(n-1)} + \dots + c_n' \cdot y_n^{(n-1)} = \dfrac{b(t)}{a_0(t)} \end{cases}$$

$$x = x_0 + x_p$$

- cazul ec. dif. cu coef. ct. este cazul în care func. a_0, $a_1, \dots, a_n$ sunt fct. ct.

- în acest caz, sol. din sist. fund. de sol. se pot determina de forma $e^{\lambda t}$ sau $t^q \cdot e^{\lambda t}$

Algoritm: - se rez. ec. dată folosind polinomul caract:

$a_0 \cdot \lambda^n + a_1 \cdot \lambda^{n-1} + \ldots + a_n = 0$ → ec. caract.

Cazul 1: $\lambda_1, \ldots, \lambda_p$ cu ordinele de multiplicitate $n_1, \ldots, n_p$ =

(răd. reale). $= \sum_{i=1}^{p} n_i = n$

$e^{\lambda_1 t}, t e^{\lambda_1 t}, \ldots, t^{n_1-1} e^{\lambda_1 t}$

$\vdots$

$e^{\lambda_p t}, t e^{\lambda_p t}, \ldots, t^{n_p-1} e^{\lambda_p t}$

$=$ sist. fund. de soluții

- pt. sol. $\mathbb{C}$: $\lambda_0 = \alpha \pm i\beta$, ambele răd. de ordinul n_0

- când $\lambda \in (\mathbb{C} \setminus \mathbb{R})$ (n_0)

$$t^2 e^{\lambda_0 t} = t^2 e^{\alpha t}(\cos \beta t \pm i \sin \beta t) =$$

$$= t^2 e^{\alpha t} \cos \beta t \pm i t^2 e^{\alpha t} \sin \beta t$$

$$e^{\alpha t}\cos \beta t, te^{\alpha t}\cos \beta t, \ldots, t^{n_0-1} e^{\alpha t}\cos \beta t$$

$$e^{\alpha t}\sin \beta t, te^{\alpha t}\sin \beta t, \ldots, t^{n_0-1} e^{\alpha t}\sin \beta t$$

Metoda coeficienților nedeterminați:

- se folosește la ec. dif. lin. de ordin sup. cu coef. ct. pt determ unei soluții particulare a ec. neomogene pornind de la term. liber $(b(t))$

Cazul I: $b(t) = P_m(t)$, $\lambda = 0$ nu este răd. a ec. caracteristice.

În ac. caz, se caută o răd. part. a ec. neom. de forma $Q_m(t)$

Cazul II: $\lambda = 0$ răd. de ordinul k, $P(t) = P_m(t)$.

Se caută sol. part. de forma $t^k \cdot Q_m(t)$.

Cazul III: $b(t) = e^{\alpha \cdot t} \cdot P_m(t)$ și $\lambda = \alpha$ nu este răd. a ec. caract.

Se caută sol. part. de forma $e^{\alpha t} \cdot Q_m(t)$.

Cazul IV: $b(t) = e^{\alpha t} \cdot P_m(t)$ și $\lambda = \alpha$ răd. de ord. k a ec. caracte.

Se caută sol. part. : $t^k \cdot e^{\alpha t} \cdot Q_m(t)$

Cazul V: $b(t) = e^{\alpha t}(P_{m_1}(t) \cdot \cos \beta t + P_{m_2}(t) \cdot \sin \beta t)$

dacă $\lambda = \alpha \pm i\beta$ nu este răd. ec. caract., se caută sol. part.

$e^{\alpha t}(Q_m(t) \cdot \cos \beta t + \underbrace{Q_m^*(t)}_{\text{alt } Q} \sin \beta t)$, $m = \max(m_1 \text{ și } m_2)$

Cazul VI: $b(t) = e^{\alpha t}(P_{m_1}(t)\cos\beta t + P_{m_2}(t)\sin\beta t)$

$\lambda = \alpha \pm i\beta$ rad. de ord. k.

Se caută sol. part. $t^k e^{\alpha t}(Q_m(t)\cos\beta t + Q_m^*(t)\cdot\sin\beta t)$, $m = \max(m_1 \text{ și } m_2)$.

Ecuații diferențiale de tip Euler: (cu coef. nect.)

$$a_0 x^n y^{(n)} + a_1 x^{n-1} y^{(n-1)} + \dots + a_n x y = b(t)$$

$$\boxed{x = e^t}$$

$$y' = \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{dy}{dt} \cdot \frac{1}{\frac{dx}{dt}} = \dot{y} \cdot \frac{1}{e^t}$$

$$x \cdot y' = e^t \cdot \dot{y} \frac{1}{e^t} = \dot{y}$$

$$y'' = \frac{d(y')}{dx} = \frac{d(y')}{dt} \cdot \frac{dt}{dx} = \frac{d(y')}{dt} \cdot \frac{1}{\frac{dx}{dt}} =$$

$$= \frac{\ddot{y} \cdot e^t - \dot{y} \cdot e^t}{e^{2t}} \cdot \frac{1}{e^t} = \frac{\ddot{y} - \dot{y}}{e^{2t}}$$

$$x^2 y'' = e^{2t} \cdot \frac{\ddot{y} - \dot{y}}{e^{2t}} = \ddot{y} - \dot{y}$$

$$y''' = \frac{d(y'')}{dx} = \frac{d(y'')}{dt} \cdot \frac{1}{\frac{dx}{dt}} = \frac{(\dddot{y} - \ddot{y})\, e^{2t} - (\ddot{y} - \dot{y})\, 2e^{2t}}{e^{4t}} \cdot \frac{1}{e^t}$$

$$= \frac{\dddot{y} - \ddot{y} - 2\ddot{y} + 2\dot{y}}{e^{3t}} = \frac{\dddot{y} - 3\ddot{y} + 2\dot{y}}{e^{3t}}$$

$$x^3 y''' = e^{3t} \cdot \frac{\dddot{y} - 3\ddot{y} + 2\dot{y}}{e^{3t}} = \dddot{y} - 3\ddot{y} + 2\dot{y}$$

Pt. red. ordinului ec. lin. de ord. sup. aceem următ. tehnică de lucru: - dacă cun. o s.p. a ec. neom. $y_p(x)$ prin calc. $y(x) = z(x) + y_p(x)$ se obț. o ec. omog. cu nec. z.

.....

Dacă la o ec. omog. se cun. s.p. $y_p(x)$, prin $y(x) = z(x) \cdot y_p(x)$ se obț. o ec. dif. având ordinul cu 1 un. mai mică decât ec. inițială.